iQOO HACKATHON 2026 · IDEA SUBMISSION

RasQ

Turning a modern smartphone into a Raspberry Pi-like compute module

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SBCs are costly; phones sit underused

The SBC problem

Raspberry Pi-class hardware is increasingly expensive and hard to source reliably, especially for students, makers and small teams. Camera, IMU, GPS and wireless modules cost extra.

The untapped smartphone

Modern phones already have powerful CPU/GPU/NPU, storage, cameras, Wi-Fi, Bluetooth and many sensors, yet much of it goes unused.

The missing piece is hardware access: phones have the compute and sensors, but no developer-friendly physical interfaces.

The smartphone becomes the compute module



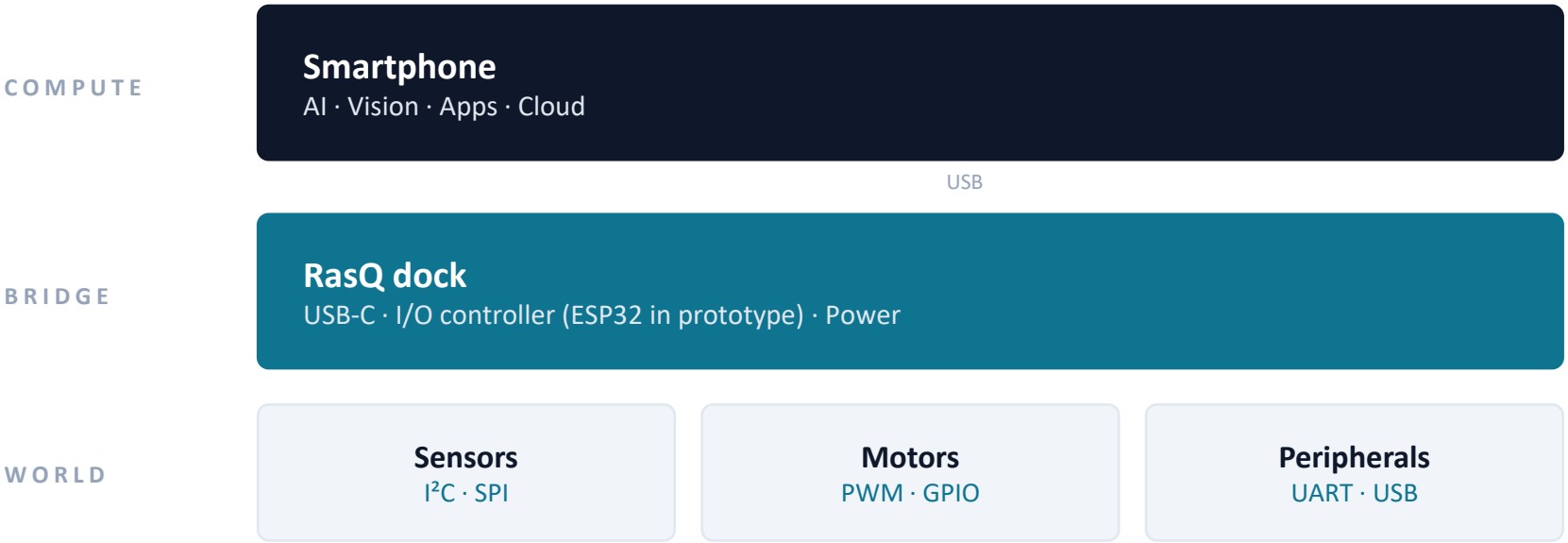
GPIO · I²C · SPI · UART · PWM · ADC

For students, developers, robotics teams, researchers and makers. No motherboard modification.

A new SBC architecture, not a Pi clone

	Raspberry Pi	RasQ
Compute	Dedicated SBC	Existing phone (CPU/GPU/NPU)
Cost	Rising cost, supply issues	Reuses the phone you own
Camera	Separate module	Built in
Connectivity	Wi-Fi/BT variants	Wi-Fi, Bluetooth, 5G built in
Sensors	Add-on boards	IMU, GPS and more built in
GPIO	On-board header	Dock-provided GPIO

The phone thinks. The microcontroller controls.



A clear stack from app to actuator

01 **Application**

02 **RasQ SDK**

03 **Hardware & USB manager**

04 **RasQ protocol over USB**

05 **I/O controller firmware**

```
GPIO_WRITE|17|1
  → OK|GPIO_WRITE|17
I2C_READ|0x76|0xF7|6
  → I2C_DATA|…
```

The Android app detects the dock, talks to the I/O controller, exposes phone sensor data, shows real-time hardware data and provides APIs for external devices.

Simple code in, on-device intelligence out

```
from rasq import GPIO, Sensor, Motor

GPIO.write(17, 1)
temp = Sensor.read("temperature")
Motor.set_speed(1, 70)
```

An SBC-style developer experience on a phone.

- Camera / sensors
- On-device AI
- Decision logic
- USB to I/O controller
- Actuator

One complete sensing-to-actuation loop

01

Hardware

USB-C prototype dock with ESP32 to validate the architecture, sensor, servo, motor driver, LEDs.

02

Software

Android app that detects the dock, reads sensors, controls GPIO and servo, shows live data.

03

AI

Camera and sensor input processed on-device, driving an external actuator.

Camera/sensors → AI → decision → USB → I/O controller → actuator

From proof of concept to dedicated dock

01

Proof of concept

USB link and protocol

02

Interfaces

GPIO, I²C, SPI, UART, PWM, ADC

03

Software

App, SDK, dashboard

04

AI

Vision to hardware control

05

Dedicated dock

Replaces ESP32 prototype

06

Final prototype

Enclosure, docs, demos

Technical challenges, already answered

Phones do not expose GPIO	→	Dedicated I/O controller (ESP32 to validate)
USB communication	→	Custom USB serial protocol
Power draw	→	Externally powered dock
Real-time control	→	Microcontroller handles low-level tasks
AI workloads	→	Phone CPU/GPU/NPU
Hardware compatibility	→	Modular interface layer

No motherboard modification. Mature USB, standard buses and existing AI frameworks.

A team that has built and shipped

Joshua Xavier Thomas

AI · Software · Product

NASA Space Apps 2025 nominee

Smart parking, accident alerts

IEEE RAS grant (USD 2,000)

Vyshak Menon

Research · Computer Vision

OpenCV species identification

Automated reporting pipeline

Game development

Nekhal James

Hardware · Robotics · Embedded

RAG-based autonomous robot

Nexus Wars, 1st prize

Mini ATV, 3rd prize

Affordable edge computing, already in your pocket.

RasQ makes the smartphone a removable compute-and-sensing module for robotics, IoT, automation and edge AI.